



A QR Code link to your video submission

Abstract

I will be using machine learning to read through movie scripts for certain scenes and flag areas containing things such as strong language, potential violent content, or potential sexual content along with a recommended age rating. It will be learning the context of the language used and age rating classification criteria to do this. My program will take inputs of excerpts from movie scripts, and I will be using scraping tools to get movie scripts with age ratings attached.

Aims and objectives:

The aims of this project are to make a functional machine learning system that will accurately highlight explicit content in provided text and recommend an age rating following UK standards

- e.g. The project objectives are:
- My program will identify explicit content in inputted text
 - My program will highlight the flagged part(s) of the text and print it back out
 - My program will provide a recommended age rating for this content following UK standards
 - My program will provide reasoning as to why the stated age rating was selected
 - I will be using NLP Python libraries to teach my program text analysis
 - I will be using recurrent neural networks for contextual understanding, as well as BERT/GPT
 - I will be using online public access websites with movie scripts for training/testing data and will use IMDb to attach age ratings to these
 - I will be referring to UK rating board guidelines to teach my program what to look for regarding different age classifications

Research:

For my research, I looked into which software to use and ultimately decided on Python. I then looked into the different aspect of NLP such as text preprocessing and the different methods used to accomplish these. I found some articles on PyTorch and TensorFlow to understand how machine learning in NLP worked and get the base concepts. I did heavy research into methods such as Bag Of Words and Topic Modelling. While I was initially interested in using Bag Of Words, I found that the “standard bag-of-words model does not capture the effect of negation and the contextual polarity efficiently” (Kasthuriarachchy, De Zoysa, Premaratne, 2014), which lead to me choosing to use topic modelling, as it is much closer aligned to the details of this project

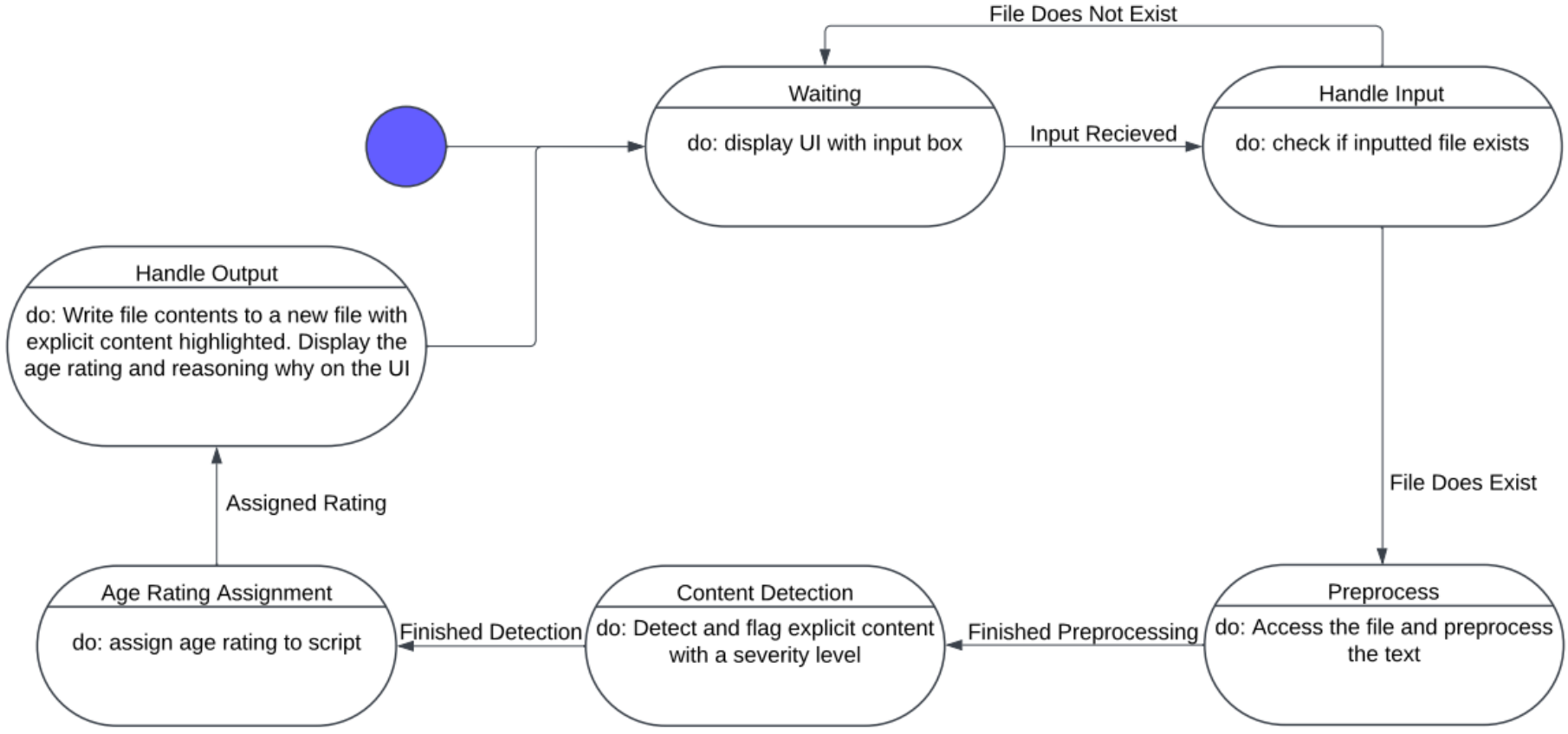
I then moved on to looking at deeper neural networks with pre-trained models to improve these algorithms, such as BERT and GPT. I found research papers such as Mohamed and Ha (2020) which were similar in nature to my project and all used BERT along with testing other methods. While they were several limitations to BERT addressed, I decided that it seemed the most fitting for my project in the end after several considerations.

Key requirements:

Functional	Non-Functional
Handle inputs	Process text and provide output within a reasonable timeframe
Detect explicit content in text	Ensure high accuracy in assigning age ratings
Identify and classify offensive or inappropriate content based on severity	Minimize false positives and negatives
Highlight identified content	Be able to handle large texts efficiently
Analyse the text to provide a recommended age rating based on UK standards	Provide a user-friendly interface with clear sections for input and output
Provide reasons for the assigned ratings	Avoid retaining user data and ensure privacy
Use a pre-trained machine learning model to detect nuanced context	Align recommendations with official UK rating standards
	Avoid ethical or legal concerns by providing transparent and explainable results
	Allow integration of additional content detection models in the future

Design, Implementation & Testing

State *Diagram*



Implementation *artefacts*

Screenshot of code writing to file:

```
Enter the name of the file you wish to check: invalid input
Error: The file 'invalid input' was not found.
Enter the name of the file you wish to check: movie scripts\Erasing Eden (2016).txt
Profanity check complete. Results saved to '<_io.TextIOWrapper name='checked_movies\\Erasing Eden (2016).txt' mode='w' encoding='utf-8'>'.

```

Screenshot of created file showing profanity detection:

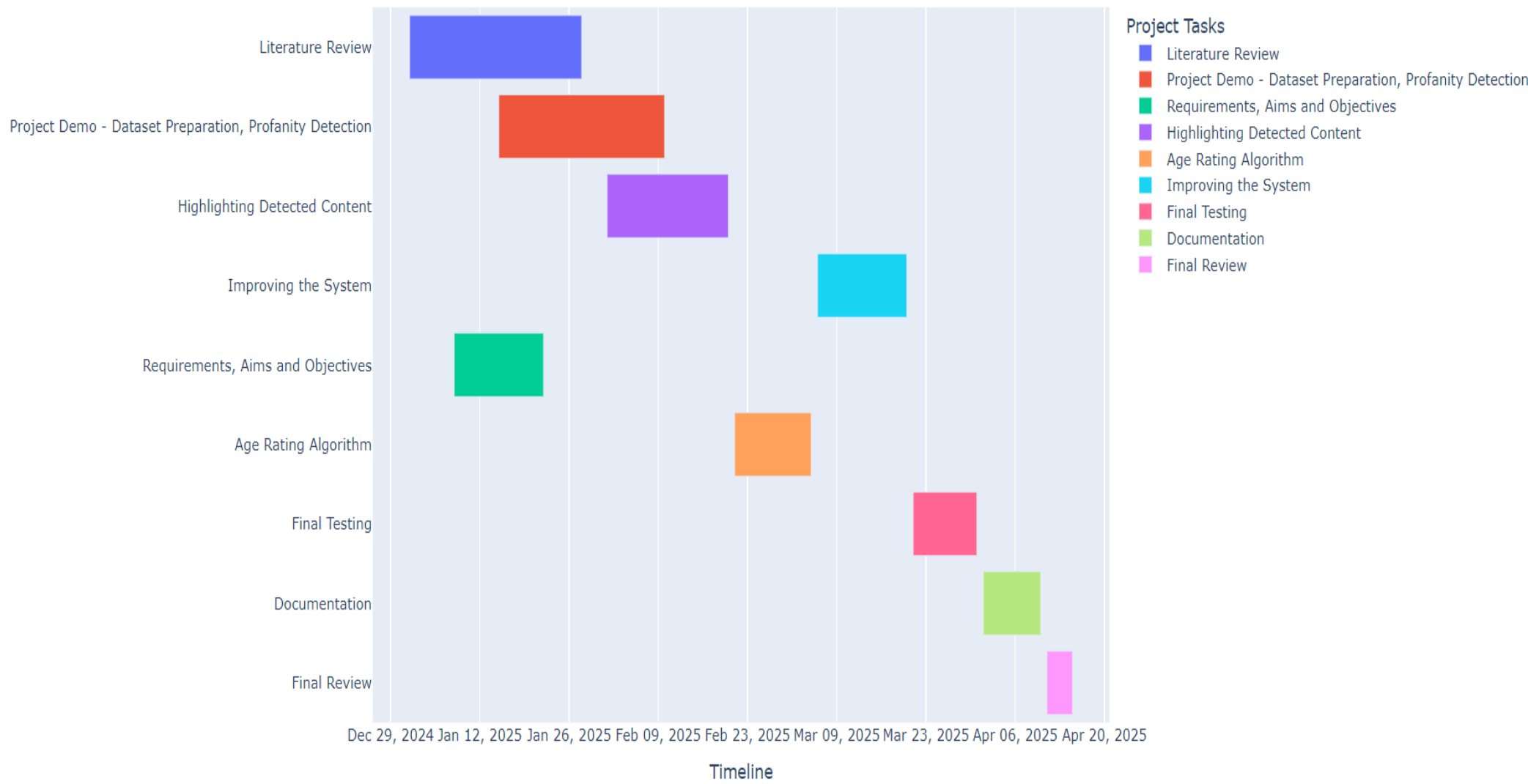
```
and that I deserve to be happy.
It's okay.
It's okay to have
**[PROFANITY DETECTED]** good shit happen.
It's okay to be happy, you know.
- No.
- Eden, stand up.

```

Testing

For my testing strategy, I will be using a train and test set based on my dataset. This should be easy to accomplish with BERT. I have added the official age ratings to the movie scripts I have downloaded, and I will be training my program on these, and then testing the rest of the set against the training set to receive a percentage accuracy. I can then add more layers to my program based on these results and receive an accuracy each time. This will only cover the broader scope of getting accurate age ratings though, and I will likely have to manually verify a few scripts each time by printing out the results from just one or two and checking if the reasoning and flagged content is correct, adjusting the code if not

Planning and Management:



References:

Kasthuriarachchy, B.H., De Zoysa, K. and Premaratne, H.L. (2014) *Enhanced bag-of-words model for phrase-level sentiment analysis*. In: *2014 14th International Conference on Advances in ICT for Emerging Regions (ICTer)*2014/12/10-2014/12/13. IEEE, pp. 210–214.

Mohamed, E. and Ha, L. (2020) A first dataset for film age appropriateness investigation. Calzolari, N. et al., eds. *International Conference on Language Resources and Evaluation* [online]. pp. 1311–1317. Available from: <https://aclanthology.org/2020.lrec-1.164/>.